

2.1.2) Material selection

Comsol allows laser welding simulation of welding process with commercial metal such as Aluminum and Cobalt. The parent metal employed in these study is Aluminum 1050 and Cobalt satellite21, which are among one of the best engineering material in automotive engineering. The dimension of the two work piece of aluminum 1050 & satellite21 welded together is 30 x 10 x 2.5 mm. The properties of these parent metals are presented below respectively;

Table 1: Material property

PROPERTY	VALUE		UNIT
Melting point	646-900	1295-1600	°C
Thermal conductivity	229	14.5-15.1	W/mK
Density	2.71	8.5	g/cm ³
Thermal expansion	24×10^{-6}	12.5×10^{-6}	K

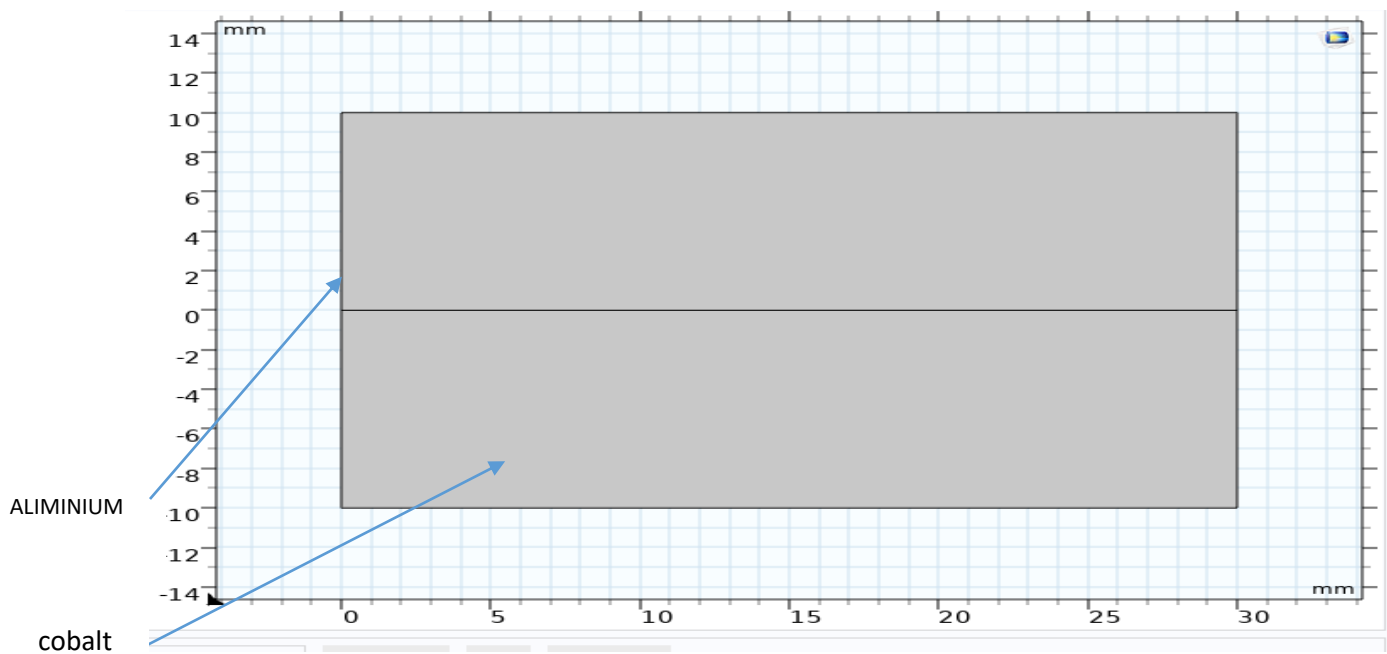


Figure 11: Aluminum sample

The process parameters which are to be optimized, were set in levels in the simulation process. Each parameter had 03 levels. The effect of these process parameters were considered for evaluation of effective response. The process parameters with their respective levels are shown Below;

Table 2: Parameters and their level

Parameters	Unit	Level 1	Level 2	Level 3
Laser power	Watt (W)	1400	1600	1800
Scan speed	mm/min	200	400	600
Focal position	mm	-1	0	1

2.3) GOVERNING EQUATION

Laser Heat source equation in the context of laser welding is primary governed by the principles of heat transfer and energy deposition.

It is important to understand that when a laser beam is focused on a material, it transfers energy that can be modelled mathematically. The three ways in which heat transfer can occur are: convection, radiation, conduction.

In our experiment conducted in Comsol, heat transfer occurs through conduction: which mostly is used in engineering to model a laser heat source. On a molecular scale thermal energy is a random motion of atoms and molecules that makeup an object, atoms are connected to each other by bond forming a lattice structure, vibration of atom through the lattice, which involves thermal energy being redistributed from area of high energy to area of low energy

The governing heat equation used in Comsol for modelling laser heat source was derived in the following way (we need to model there heat transfer);

Step 1: We can derive Q using Fourier laws (Q stand for heat transfer rate; which is the amount of energy that flows through a defined areas each second)

$$Q = -KA \frac{dT}{dx} \text{ ----- (1)}$$

where;

$\frac{dT}{dx}$ = Temperature gradient

k = thermal conductivity of the material

Step 2: heat flux equation:

$$Q_f = -K \frac{dT}{dx} \text{ ----- (2)}$$

$$Q_{x+dx} = Q_x + \frac{\partial}{\partial x}(Q_x)dx \text{ ----- (3)}$$

$$Q_{x+dx} = -k_x \frac{\partial T}{\partial x} dydz - \frac{\partial}{\partial x} [k_x \frac{\partial T}{\partial x}] dx dydz, \text{ ----- (4)}$$

where the heat flux in the x-direction:

$$Q_x = -k_x \frac{\partial T}{\partial x} dydz$$

Step 3: Deduce the heat transfer equation:

From the above, the expression of the heat flux difference in the x-direction gives;

$$Q_x - Q_{(x+dx)} = -k_x \frac{\partial T}{\partial x} dydz - \left[-k_x \frac{\partial T}{\partial x} dydz - \frac{\partial}{\partial x} \left[-k_x \frac{\partial T}{\partial x} \right] dx dydz \right] \text{ ----- (5)}$$

After simplification we have;

$$Q_x - Q_{(x+dx)} = \frac{\partial}{\partial x} \left[-k_x \frac{\partial T}{\partial x} \right] dx dydz \text{ ----- (6)}$$

Similarly in the y and z directions, we obtain the heat flux equation as;

$$Q_y - Q_{(y+dy)} = \frac{\partial}{\partial y} \left[-k_y \frac{\partial T}{\partial y} \right] dx dydz, \text{ ----- (7)}$$

$$Q_z - Q_{(z+dz)} = \frac{\partial}{\partial z} \left[-k_z \frac{\partial T}{\partial z} \right] dx dydz. \text{ ----- (8)}$$

Summing equation (6), (7), (8), leads to;

Total heat conducted into the element from all direction (G);

$$\left[\frac{\partial}{\partial x} \left[-k_x \frac{\partial T}{\partial x} \right] + \frac{\partial}{\partial y} \left[-k_y \frac{\partial T}{\partial y} \right] + \frac{\partial}{\partial z} \left[-k_z \frac{\partial T}{\partial z} \right] \right] dx dy dz \text{ ----- (9)}$$

The change in enthalpy of the element $\frac{dy}{dx}$ is defined as;

$$\left\{ \begin{array}{c} \text{Change of Enthalpy} \\ \text{in the element} \end{array} \right\} = \left\langle \begin{array}{c} \text{mass of the} \\ \text{element} \end{array} \middle| \begin{array}{c} \text{specific heat} \\ \text{of the element} \end{array} \middle| \begin{array}{c} \text{rise in temperature} \\ \text{of the element} \end{array} \right\rangle \text{ ----- (10)}$$

$$\left\{ \begin{array}{c} \text{Change of Enthalpy} \\ \text{in the element} \end{array} \right\} = (\rho \times dx dy dz) \times c \rho \frac{\partial T}{\partial t} \text{ ----- (11)}$$

The Heat generated from the inner heat source within the element Q_g is expressed as;

$$Q_g = q_g \times dx dy dz \text{ ----- (12)}$$

Making use of the first law of thermodynamic or law of conservation, the differential equation of conduction can be derived as;

$$\left\{ \begin{array}{c} \text{Quantity of heat} \\ \text{conducted to the} \\ \text{volume} \\ Q_x \end{array} \right\} + \left\{ \begin{array}{c} \text{Heat generated from} \\ \text{the inner heat source} \\ \text{of the volume} \\ Q_g \end{array} \right\} = \left\{ \begin{array}{c} \text{change of Enthalpy} \\ \text{of the volume} \\ \frac{dh}{dt} \end{array} \right\} \text{ ----- (13)}$$

Hence, the above equation can be written as;

$$Q_x = \frac{dh}{dt} + Q_x + dx \quad \text{-----} \quad (14)$$

Substitute equation (9), (11), and (12) in equation (14), we have;

$$\left[\frac{\partial}{\partial x} \left[k_x \frac{\partial T}{\partial x} \right] + \frac{\partial}{\partial y} \left[k_y \frac{\partial T}{\partial y} \right] + \frac{\partial}{\partial z} \left[k_z \frac{\partial T}{\partial z} \right] \right] dx dy dz + q_g dx dy dz = \rho c \rho \frac{\partial T}{\partial t} dx dy dz \quad \text{----} \quad (15)$$

$$\frac{\partial}{\partial x} \left[k_x \frac{\partial T}{\partial x} \right] + \frac{\partial}{\partial y} \left[k_y \frac{\partial T}{\partial y} \right] + \frac{\partial}{\partial z} \left[k_z \frac{\partial T}{\partial z} \right] + q_g = \rho c \rho \frac{\partial T}{\partial t} \quad \text{-----} \quad (16)$$

Divided by k;

$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} + \frac{q_g}{k} = \frac{\rho c \rho}{k} \frac{\partial T}{\partial t} \quad \text{-----} \quad (17)$$

$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} + \frac{q_g}{k} = \frac{1}{\alpha} \frac{\partial T}{\partial t} \quad \text{-----} \quad (18)$$

Where we divided all through by K, and ∇^2 which represents the laplace operator

$$\nabla^2 T + \frac{q_g}{k} = \frac{1}{\alpha} \frac{\partial T}{\partial t} \quad \text{-----} \quad (19)$$

It is a general three dimensional heat conduction equation in Cartesian coordinates

$$\text{Where, } \alpha = \textbf{Thermal conductivity} = \frac{k}{\rho c \rho} \quad \text{-----} \quad (20)$$

In my study during my simulation of a laser heat source on the material aluminum 1100 in Comsol, there was a parameter consider as internal heat generation which was zero ($q_g = 0$), so the final Fourier equation used for there laser heat source modelling was;

$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} = \frac{1}{\alpha} \frac{\partial T}{\partial t} \quad \text{-----} \quad (21)$$

$$\nabla^2 T = \frac{1}{\alpha} \frac{\partial T}{\partial t} \quad (\textbf{Fourier equation}) \quad \text{-----} \quad (22)$$

To model mathematically a laser heat source in Comsol, you need to describe the spatial and temporal distribution of the laser energy. Which involve mathematical models like heat transfer, Gaussian heat source model and also there Gaussian space.

2.3.1) Gaussian heat source

These is a mathematical model that describes how heat is distributed from a source with most heat concentrated at the center, like a laser beam and its take into account there beams elliptical shape, which is it shape function. There mathematical model is giving by;

$$Q_{tot} = (2 \times q_{laser} / (\pi \times \omega_0^2)) \times \exp(-2 \times r^2 / \omega_0^2) \quad \text{-----} \quad (23)$$

Where;

- Q_{tot} is the total heat flux (w/m²)
- Q_{laser} is the laser power (w)
- ω_0 is the beam waist radius (m)
- R is there radial distance from the beam center

2.3.2) Gaussian space

These is a mathematical model that describes a distribution of points in space, where the density of points decreases as you move away from the center. It is characterized by its symmetry, with highest density of points at the center and decreasing density as you move away from the center. The Gaussian heat source and Gaussian space are connected through their shared origin. Its mathematical model are;

$$r = \sqrt{((x - x_{focus})^2)} \quad \text{-----} \quad (24)$$

Where;

- x is the point you're interested in
- x_{focus} is the focus point (center) of the Gaussian distribution

2.5.1) Simulation runs according to orthogonal array (COMSOL)

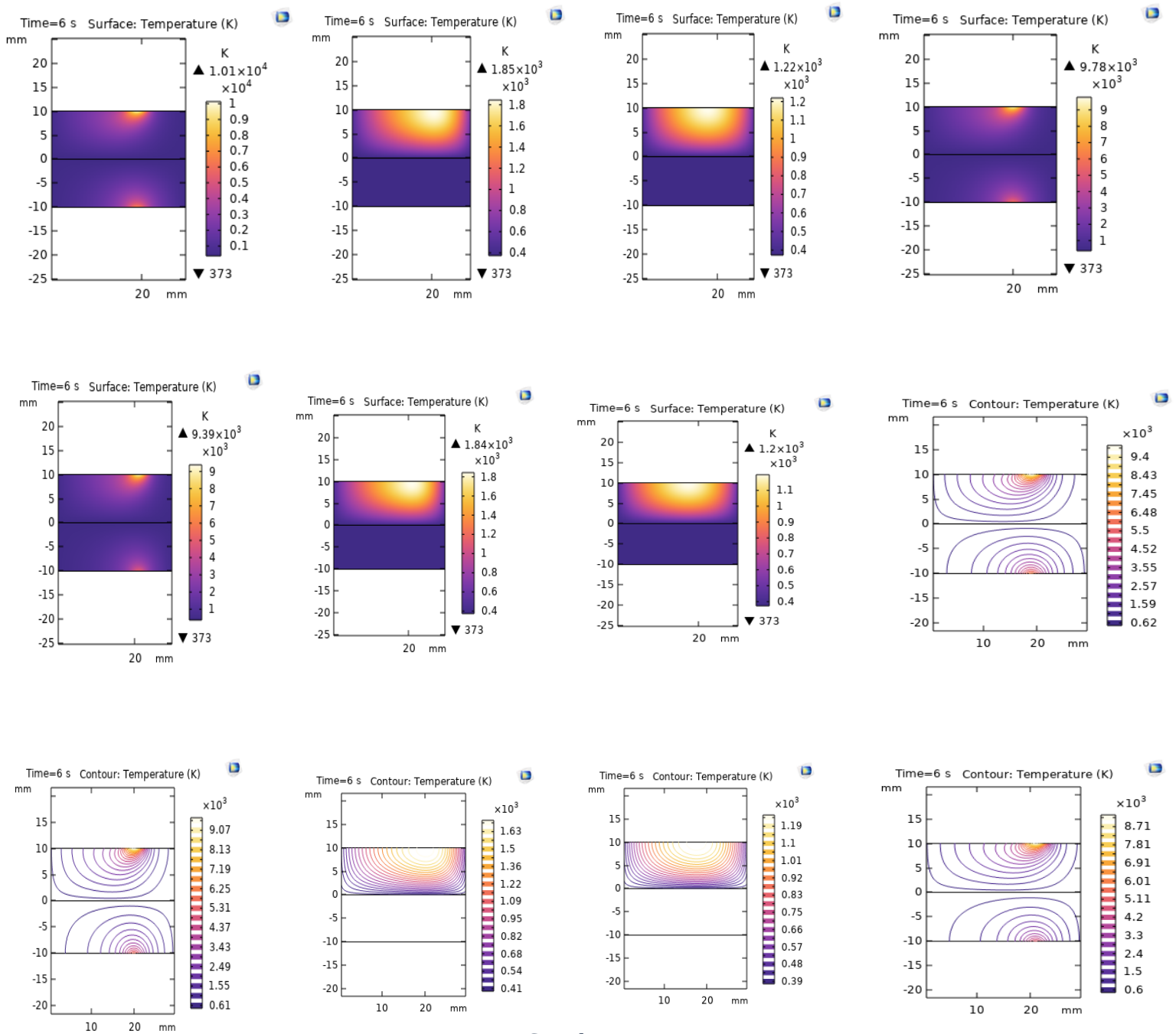


Figure 12: Simulation runs